x_1 x_2 x_3 x_4 x_5 x_6

$\begin{pmatrix}1\\1\end{pmatrix}$ $\begin{pmatrix}2\\1\end{pmatrix}$ $\begin{pmatrix}1\\2\end{pmatrix}$ $\begin{pmatrix}5\\4\end{pmatrix}$ $\begin{pmatrix}6\\5\end{pmatrix}$ $\begin{pmatrix}6\\6\end{pmatrix}$

C_1 C_1 C_1 C_2 C_2 C_2

Initial $\underline{\mathbf{W}} = \begin{pmatrix}4\\-2\\-2\end{pmatrix}$ ***Step η = 2***

C_1 positive C_2negative Zero is unaccepted

Check classes of Z_1 Z_1 Z_1

$\begin{pmatrix}0\\0\end{pmatrix}$ $\begin{pmatrix}7\\6\end{pmatrix}$ $\begin{pmatrix}3\\2\end{pmatrix}$

Ugmented Vectors (from Top)

x_1 x_2 x_3 x_4 x_5 x_6

$\begin{pmatrix}1\\1\\1\end{pmatrix}$ $\begin{pmatrix}1\\2\\1\end{pmatrix}$ $\begin{pmatrix}1\\1\\2\end{pmatrix}$ $\begin{pmatrix}1\\5\\4\end{pmatrix}$ $\begin{pmatrix}1\\6\\5\end{pmatrix}$ $\begin{pmatrix}1\\6\\6\end{pmatrix}$

C_1 C_1 C_1 C_2 C_2 C_2

Check Classes

$d(Z_1) = (1\ 0\ 0)\begin{pmatrix}14\\-4\\0\end{pmatrix} = 14$ ⇒ **Positive > 0** **Class I**

$d(Z_2) = (1\ 7\ 6)\begin{pmatrix}14\\-4\\0\end{pmatrix} = -14$ ⇒ **Negative < 0** **Class II**

$d(Z_2) = (1\ 3\ 2)\begin{pmatrix}14\\-4\\0\end{pmatrix} = -2$ ⇒ **Negative < 0** **Class II**